

MOHIT AJAYKUMAR DHABUWALA

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Summary

Cyber Forensics graduate with expertise in digital evidence acquisition and analysis, supported by strong foundational knowledge in Linux, Python, and computer architecture. Not solely reliant on forensic tools—skilled at decoding complex data and scripting custom automation workflows. Known for analytical thinking, clear and ethical reporting of sensitive data. OPT starts June 2025.

Technical Skills

- Digital Forensics & Incident Response:** Memory, Windows, Mobile, and Cloud Forensics; Digital Evidence Acquisition & Preservation; Anti-Forensics Detection; Malware Analysis; IoC Analysis (YARA); Incident Triage ; MITRE ATT&CK Framework
- Forensic Toolkit:** Proficient in Magnet AXIOM, EnCase, FTK Imager, Belkasoft X, Autopsy, Cellebrite UFED, Paraben E3, Oxygen Forensics, The Sleuth Kit (TSK), Volatility, OSForensics, Wireshark
- Operating Systems & File System Analysis:** Windows (NTFS, FAT, Event Logs, Registry), Linux (EXT4), macOS (APFS/HFS) and Android/iOS forensic artifacts, including system logs and app data.
- Programming & Automation:** Python, Bash, PowerShell, log automation (RAG), metadata extraction (Regex, SQL).
- Professional Competencies:** Analytical Reasoning, Forensic Report Writing, Chain of Custody Compliance, Time Management

Education

University of Baltimore, Baltimore, USA May 2025

M.S. in Cyber Forensics and Counterterrorism | GPA: 4.00 / 4.00

Sarvajanik College of Engineering and Technology, Surat, India June 2023

B.E. in Electronics and Communication Engineering | GPA: 3.52 / 4.00

Experience

University of Baltimore, Baltimore, USA Jan 2024 – Present

Graduate Research Assistant

- Developed an AI-driven model using Stable Diffusion to extract and enhance degraded text from picture based forensic artifacts, achieving a 30% improvement in readability and enabling efficient recovery.
- Conducted a forensic investigation on a Google Pixel 3 using Linux command-line, verifying findings with Magnet Axiom and authored comprehensive forensic reports and executive summaries, detailing investigation with actionable recommendations.

Hexxa Geo Systems Integrators Pvt. Ltd., Surat, India Feb 2023 – Apr 2023

Technical Support Engineer

- Improved incident detection and response times by 15% by optimizing the network integration of physical security systems, ensuring faster identification and mitigation of security events.
- Enhanced security monitoring and proactively maintained system uptime of 80% through continuous diagnostics and troubleshooting of networked security components, including firewalls, and endpoint protection, identifying and mitigating an average of 5 vulnerabilities per system.

eInfochips - An Arrow Company, Ahmedabad, India Jun 2022 – Jul 2022

ASIC Design Engineer

- Engineered a 2-stage CMOS Op-Amp in 180nm technology, achieving a 95% gain increase, significantly improving signal reconstruction capabilities for high-performance forensic hardware analysis.
- Reduced chip power consumption by 15% using strategic design enhancements, ensuring stable, low-power operation in portable forensic tools, minimizing data loss during on-site electronic evidence acquisition.

Projects and Research

AI-Powered Text Recovery from Distorted Forensic Images

- Applied Stable Diffusion-based AI algorithms to enhance text visibility in degraded photo evidences, enhancing text readability by 30% and expediting digital investigations by automating evidence processing.
- Processed digital evidence from distorted image artifacts using Python, OpenCV, and deep learning frameworks (TensorFlow, PyTorch) to extract critical forensic information, improving digital evidence recovery speed by 60%.

Cross-Platform Digital Forensics Case Studies

- Conducted comprehensive forensic examinations across mobile (Android, iOS) and computer systems, utilizing tools like FTK Imager, Magnet Axiom, and Autopsy to extract and analyze critical artifacts from diverse digital sources.
- Successfully reconstructed user activities and investigated potential security breaches by analyzing system logs, mobile app data, and network traffic, demonstrating proficiency in data recovery and forensic analysis across multiple platforms.

Certifications

- Advanced YARA & Sigma rules for Digital Forensics** - Belkasoft (Feb 2025)
- Windows Forensic Certification** - Belkasoft (Jan 2025)
- Magnet Certified Forensic Examiner (MCFE)** - Magnet Forensics (Dec 2024)
- Paraben Mobile and Computer Operator (PMO/PCO)** - Paraben Corporation (Sept 2024)